

# Torsion Testing of 4140 Steel and 6061 Aluminum

Shear modulus, Poisson's ratio, and failure mode of cylindrical specimens under torsion

|           |                                                                    |            |                                                                                        |
|-----------|--------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------|
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| Course    | MECH 237A Strength of Materials Lab, NJIT                          | Instructor | Prof. Milano · TA: Oscar Poudel                                                        |
| Team      | Kacper Czarnecki, Ashlee Hercules, Luis Ramirez, Jackson Robillard | My role    | Wrote the procedure and data acquisition section; individual discussion and conclusion |
| Equipment | Torsion testing machine, data acquisition system, vernier caliper  |            |                                                                                        |

## PROJECT SUMMARY

This experiment examined how metals behave when twisted. Cylindrical specimens of 4140 steel and 6061 aluminum were loaded in a torsion testing machine until failure while torque and angle of twist were recorded. Shear stress and shear strain were calculated from the data, and the shear modulus  $G$  was taken as the slope of the linear elastic region. The measured shear modulus was 78.1 GPa for steel (2.4% from the 80 GPa reference) and 24.5 GPa for aluminum (6.2% from 26 GPa). Poisson's ratio calculated from  $E$  and  $G$  was 0.31 for steel and 0.41 for aluminum. Both specimens fractured on a plane normal to the specimen axis after extensive twisting, the signature of ductile torsional failure.

## 1. Introduction & Theory

Torsion occurs when an external torque twists a member, producing shear stress and shear strain. Understanding torsional behavior is essential for designing shafts, fasteners, and other components that carry twisting loads. The objective of this experiment was to measure the torque–twist response of two metals, determine their shear modulus and Poisson's ratio, and observe how they fail under shear.

The key relationships are the torsion formula  $\tau = Tr/J$ , the shear strain  $\gamma = r\phi/L$ , and the angle of twist  $\phi = TL/GJ$ , where  $T$  is torque,  $r$  the radius,  $J$  the polar moment of inertia,  $L$  the length, and  $G$  the shear modulus. In the elastic region, Hooke's law in shear gives  $\tau = G\gamma$ . Finally, for an isotropic material the shear modulus, elastic modulus, and Poisson's ratio are linked by  $G = E / 2(1 + \nu)$ , so  $\nu = E/2G - 1$ .

## 2. Procedure

**Equipment.** A vernier caliper was used to measure each specimen's diameter and length. A permanent marker was used to draw a straight reference line along the length of each specimen, so the twist could be seen clearly. The torsion testing machine applied a controlled torque until failure, while a data acquisition system recorded torque and angular displacement. A ruler was used for additional length measurements and to sketch the failure mode.

**Data acquisition.** After measuring and marking each specimen, the team aligned it in the testing machine and zeroed the rotation. Torque was increased gradually and in a controlled way until the specimen failed, and the failure torque was recorded. The failure surface was then sketched and photographed. From the recorded data, the team plotted torque versus rotation and shear stress versus shear strain for the full test, then isolated the linear elastic region to determine  $G$ . Poisson's ratio was calculated from  $G$  and the reference elastic modulus, and all results were compared with published values.

## 3. Results

**Table 1. Specimen dimensions and measured torsional properties**

| Parameter                             | 4140 Steel | 6061 Aluminum |
|---------------------------------------|------------|---------------|
| Gauge length $L$ (mm)                 | 57.41      | 57.90         |
| Diameter $d$ (mm)                     | 9.50       | 9.54          |
| Polar moment $J$ (mm <sup>4</sup> )   | 799.64     | 813.19        |
| Shear modulus $G$ , measured (GPa)    | 78.105     | 24.477        |
| Shear modulus $G$ , reference (GPa)   | 80         | 26            |
| Error in $G$                          | 2.43%      | 6.22%         |
| Elastic modulus $E$ , reference (GPa) | 205        | 68.9          |
| Poisson's ratio $\nu = E/2G - 1$      | 0.31       | 0.41          |
| Poisson's ratio, reference            | 0.29       | 0.33          |
| Error in $\nu$                        | 7.15%      | 19.0%         |

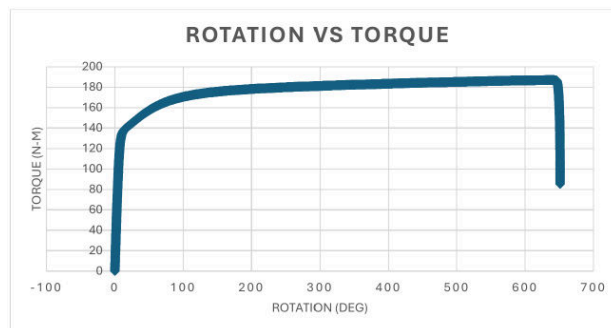


Figure 1. Steel: torque vs. rotation

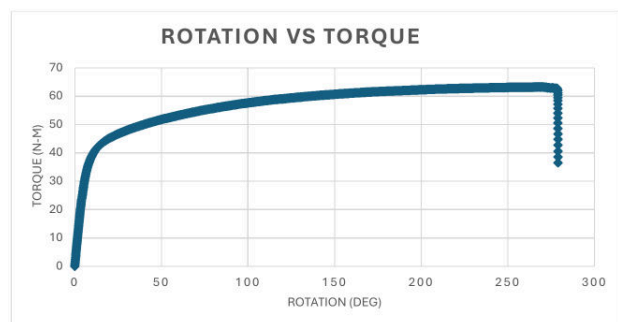


Figure 2. Aluminum: torque vs. rotation

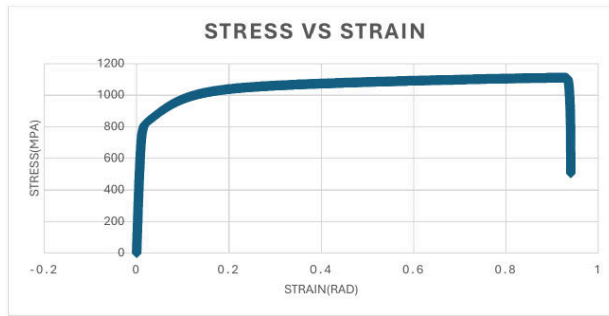


Figure 3. Steel: shear stress vs. shear strain

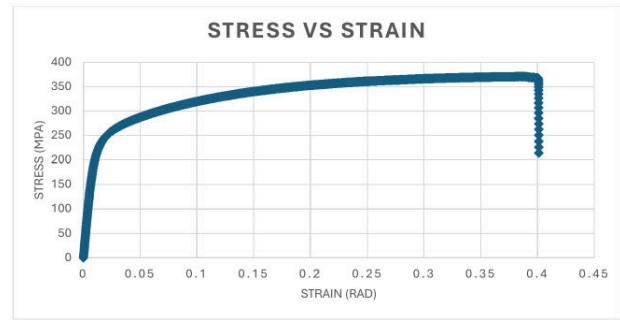


Figure 4. Aluminum: shear stress vs. shear strain

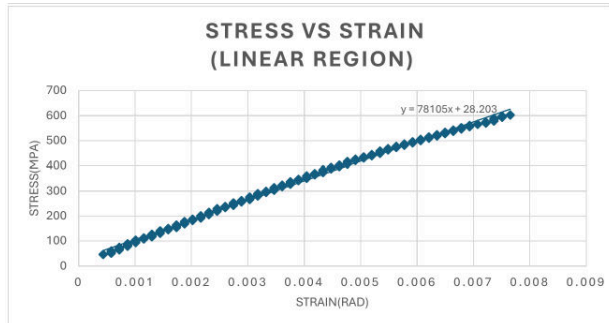


Figure 5. Steel linear region (slope =  $G = 78,105$  MPa)

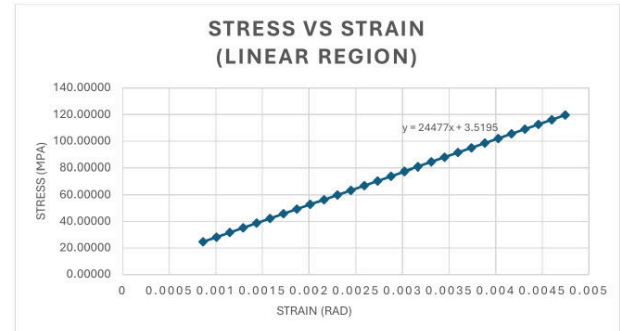


Figure 6. Aluminum linear region (slope =  $G = 24,477$  MPa)



Figure 7. Both specimens after failure

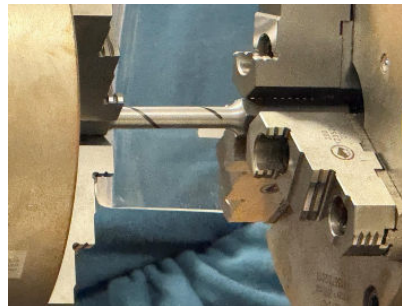


Figure 8. Steel specimen under torsional load

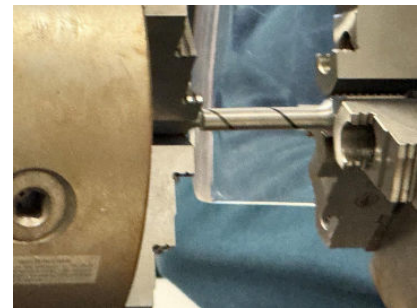


Figure 9. Steel specimen after failure

## 4. Discussion

**Shear modulus.** The experimental shear moduli agreed closely with reference values: 78.1 GPa for steel (2.4% error) and 24.5 GPa for aluminum (6.2% error). This low error shows the method is reliable, and it confirms that in the elastic region shear stress is proportional to shear strain, as Hooke's law in shear predicts. The steel carried a peak torque of roughly 185 N·m and twisted about 650° before failing, while the aluminum peaked near 63 N·m after about 280°.

**Poisson's ratio.** The steel value (0.31) was within about 7% of its reference, but the aluminum value (0.41) was 19% high. Because  $\nu$  is calculated from  $E/2G - 1$ , it amplifies small errors in  $G$ : the 6% error in the aluminum shear modulus becomes a much larger error in Poisson's ratio.

**Elastic to plastic transition.** Both materials behaved linearly at first, then entered a plastic region past the yield point where stress and strain were no longer proportional. The reference lines drawn on the specimens spiraled permanently around the specimen after testing, clear visual evidence of plastic deformation.

**Failure mode.** Both specimens twisted extensively and then fractured on a plane normal to the axis of twist. This is the characteristic failure of a ductile material, which fails in shear on the plane of maximum shear stress. A brittle material would instead fail suddenly along a 45° helix, the plane of maximum tensile stress. This matches the ductile necking observed in the tension test.

**Sources of error.** Small differences from reference values may come from variation in the material, measurement error in the specimen dimensions, and friction or compliance in the testing equipment. These effects clearly did not compromise the results significantly.

## 5. Conclusion

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The experiment met its objectives. Shear moduli of 78.1 GPa for steel and 24.5 GPa for aluminum were within 2.4% and 6.2% of reference values, and Poisson's ratio was calculated for both materials. Both metals behaved linearly up to yield and then deformed plastically, and both failed in a ductile manner after substantial twisting. These properties are essential for designing shafts, beams, and fasteners that carry torsional loads.

## References

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Ciesla, M. (2017). *Strength of materials laboratory manual*. New Jersey Institute of Technology.

Beer, F. P., Johnston, E. R., Jr., DeWolf, J. T., & Mazurek, D. F. (2020). *Mechanics of materials* (8th ed.). McGraw Hill.

Thorat, S. (2018, March 14). *Elastic constants and their relationship* – Strength of material.